CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 5: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smallest number divisible by 6, 8 and 12 is: 1

(a) 24 **(b)** 12 **(c)** 48 **(d)** 30

2. The smallest number to be added to 6713 to make it divisible by 11 is: 1

(a) 9 **(b)** 0 **(c)** 8 **(d)** 10

3. The smallest 4-digit number divisible by 12 is: 1

(a) 1020 **(b)** 1000 **(c)** 1008 **(d)** 996

4. The value of 92 - 29 is: 1

(a) 54 **(b)** 63 **(c)** 72 **(d)** 153

5. The digit that replaces * in 2*64 so that the number is divisible by 11 is: 1

(a) 0 **(b)** 7 **(c)** 5 **(d)** 4

6. The smallest digit that replaces * in 46*0 so that the number is divisible by 6 is: 1

(a) 7 **(b)** 3 **(c)** 2 **(d)** 4

7. One step of the Kaprekar routine applied to 7714 gives: 1

(a) 6165 **(b)** 6282 **(c)** 6264 **(d)** 6363

8. The smallest number divisible by 9, 12 and 15 is: 1

(a) 189 **(b)** 360 **(c)** 90 **(d)** 180

9. The smallest number to be added to 5207 to make it divisible by 6 is: 1

(a) 3 **(b)** 2 **(c)** 4 **(d)** 1

10. The smallest 4-digit number divisible by 15 is: 1

(a) 1005 **(b)** 1020 **(c)** 1000 **(d)** 990

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Check whether 7248 is divisible by 12 using the rules for 3 and 4. 2

12. Find 753 - 357 for the 3-digit number 753 and check whether it is divisible by 99. 2

13. Find the smallest number to be added to 6094 to make it divisible by 12. 2

14. Check whether 4950 is divisible by 12 using the rules for 3 and 4. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the smallest 4-digit number divisible by both 16 and 24. 3

16. Find the digit * so that 8*47 is divisible by 11. Show your working. 3

17. Express 111 as the sum of three consecutive natural numbers. 3

OR

Find the smallest 4-digit number divisible by both 18 and 27. 3

18. Find the digit * so that 5*82 is divisible by 11. Show your working. 3

OR

Express 150 as the sum of three consecutive natural numbers. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Apply the Kaprekar routine to 3187 until you reach 6174. Write every step and state the number of steps. 5

OR

Find the smallest number which leaves remainder 5 when divided by 9, 12 and 18. 5

20. Apply the Kaprekar routine to 5942 until you reach 6174. Write every step and state the number of steps. 5

OR

Find the smallest number which leaves remainder 7 when divided by 8, 12 and 20. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND